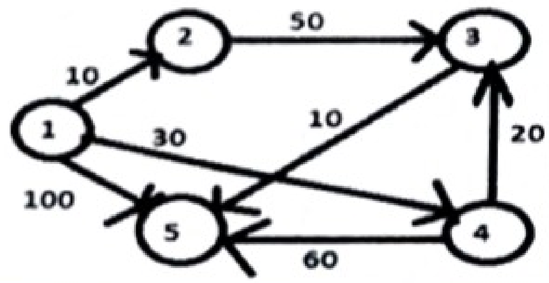
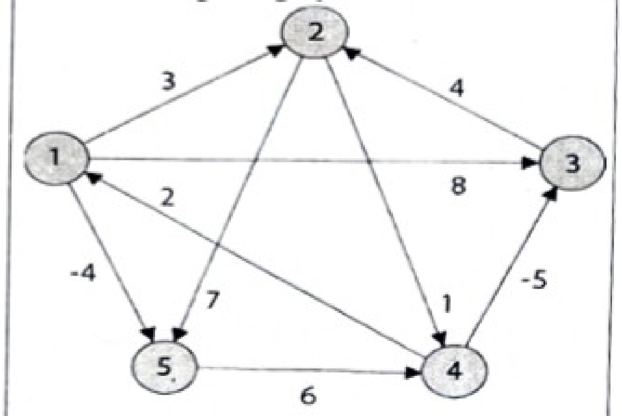
Student Name: University Roll No.:	Printed Pages: 02
School of Engineerin Second Theory Sessional Ex ination Even Semester (AS: 202 25) B. Tech: CSE/CSE-AI/CSE-IOTBC, CSE-CCML	Year: 3| |Semester :VI|
Course Title: Design & Analysis of Algorithm Course Code: BCS3602 [CSE-AI] BAI3602 [CSE, CSE-CCML, CSE-IOTBC]	Max Marks: 30 Time: 1*hrs*

Instructions: 1-Mention any assumptions made.
2-Notations have usual meaning.

SECTION 'A' Q.N.1. Attempt all parts of the following:		Course Objective	Marks
a)	Compare Greedy Programming, Dynam programming & Divide & conquer.	CO3	1
b)	Define Single source shortest path,	CO3	1
c)	Give one difference between BFS and DFS	CO4	1
d)	Define P,NP and NP-complete in decisi problems	CO4	1
e)	What do you mean by Approximation algo	CO4	1

SECTION 'B' Q.N.2. Attempt any two parts of th following:		Course Objective	Marks
a)	Describe the Dynamic 0/1 Knapsac Problem. Find an optimal solution for th dynamic programming 0/1 knapsa instance for n=3, m=6, profits are (p1, p2, p3) = (1,2,5), weights are (w1,w2,w3)=(2,3,4).	CO3	7.5
b)	Define backtracking ? Solve N-Queens problem for n=4 showing backtracking steps.	CO3	7.5
c)	Explain Rabin Karp string matching Algorithm. Simulate Rabin-Karp string matching for pattern "abc" in text "abcabcabc" with q=101.	CO4	7.5

SECTION 'C' Q.N.3. Attempt any one parts of the following:		Course Objective	Marks
a)	Find the shortest path in the below graph from the source vertex 1 to all other vertices by using **Dijkstra's algorithm.**  	CO4	10
b)	Explain Floyd warshell with an algorithm Find out all pair shortest path using Floyd warshell for given graph:  	CO5	10